

KLH UNIVERSITY

Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester(2025-26)
Faculty Name	Ms. Md Razia
Student Name	Adunuri Bhavana Sri
Roll Number	2520030232
Branch	CSE
Course Outcome	3

Case Study Topic: HealthNet – Intelligent Hospital Resource Management System using Graph Traversal (BFS)

Work Done Summary:

- Implemented a graph-based network routing model in Java using an adjacency list to connect medical units, emergency services, and care centers.
- Mapped hospitals, ambulances, and supply points as interconnected vertex configurations linked by weighted paths.
- Utilized Breadth First Search (BFS) graph traversals to evaluate resource routing pathways level-by-level, tracking shortest transfer links.
- Integrated dynamic verification components to discover node disconnections, manage vehicle distributions, and check facility network data rules.

Implementation Details:

- **1. Constructed Graph framework class** with adjacent list representations to facilitate high-throughput network pathing configurations.
- **2. addEdge()** creates structural associations between regional emergency medical terminals and health hubs.
- **3. BFS(startNode)** evaluates localized network distance tiers step-by-step using an iterative Queue tracking algorithm.

- **4. Visited indicator arrays** manage state checks to completely prevent infinite recursive tracking over cyclic layout configurations.
- **5. Main execution engine** creates the institutional health graph, initializes BFS lookups, and validates routing paths.

Result: Screenshots/Output

```
PS D:\JavaPrograms> javac HealthNetGraphBFS.java
PS D:\JavaPrograms> java HealthNetGraphBFS

HealthNet - Integrated Hospital Network Connectivity Map:
Hub 0 (City Central ER) <-> Hub 1 (North Care Outpost)
Hub 0 (City Central ER) <-> Hub 2 (West Wing Trauma Center)
Hub 1 (North Care Outpost) <-> Hub 3 (Regional Diagnostics Unit)
Hub 1 (North Care Outpost) <-> Hub 4 (Specialty Pediatric Ward)
Hub 2 (West Wing Trauma Center) <-> Hub 5 (Satellite Blood Bank)

Initializing Network Routing Audit via BFS Traversal:
Verified Hub Vertex: 0
Verified Hub Vertex: 1
Verified Hub Vertex: 2
Verified Hub Vertex: 3
Verified Hub Vertex: 4
Verified Hub Vertex: 5

Network Routing Assessment: All elements connected. 0 pathway bottlenecks found.
```

Time Complexity & Empirical Analysis:

- BFS Network Traversal Timeline: $O(V + E)$ where V maps nodes and E details pathways.
- Network Model Allocation Build: $O(E)$ edge instantiation timeline.
- BFS ensures ideal level-by-level scanning, making it perfect for rapid medical transit networks and hospital unit routing analyses.

GitHub Repository Link:

<https://github.com/bhavana-sri25/Sketch-DSA-CO3>

Student Signature

Faculty Signature